Kantipur City College

Department of Computer Engineering

Semester IV/ Probability and Statistics

Lab Sheet No.3

Title: Computing Pearson's correlation coefficient using EXCEL.

Objectives: i) Draw scatter diagram.

ii) To compute correlation coefficient and interpretation.

Theory:

Correlation coefficient (r) = $\frac{n\sum xy-\sum x\sum y}{\sqrt{n\sum x^2-(\sum x)^2}\sqrt{n\sum y^2-(\sum y)^2}}$

Lab work:

The following data is the daily rainfall and particulate level in Kathmandu valley,

Daily rainfall (X)	4.1	4.3	5.7	5.4	5.9	5	3.6	1.9	7.3
Particulate level(Y)	122	117	112	114	110	114	128	137	104

i) Draw scatter diagram.

ii) Compute Karl Pearson's correlation coefficient.

Using EXCEL:

i)Scatter diagram is plotted by using following steps.

- i)Input data in EXCEL:

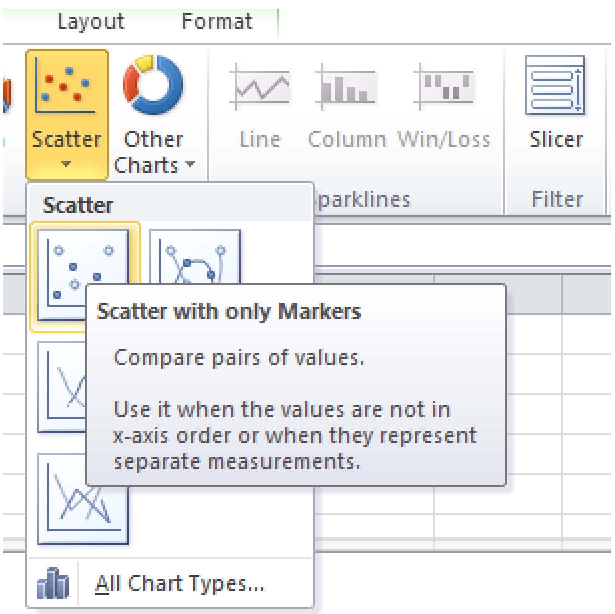
  X: Cell A2:A10

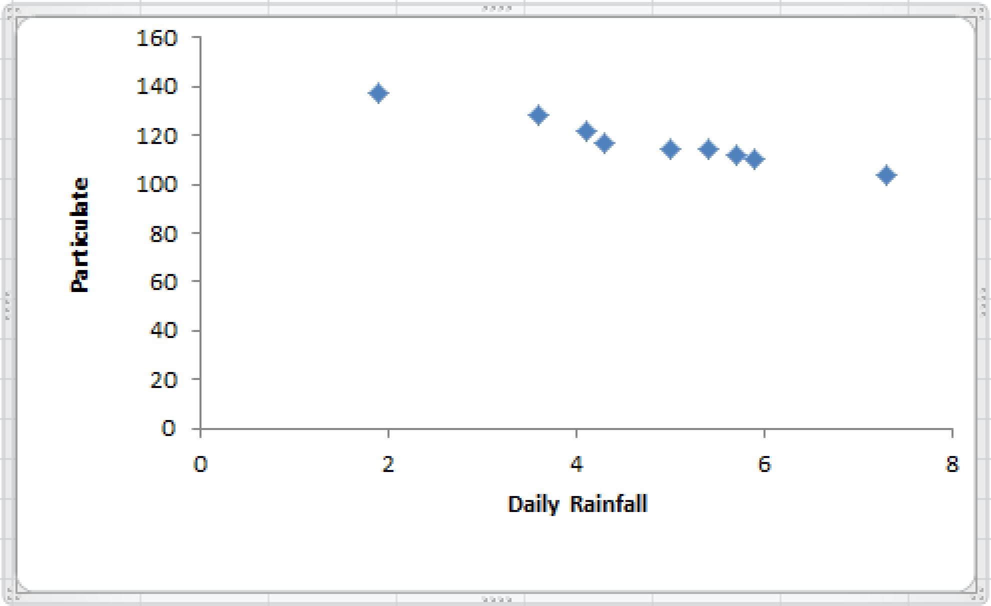
  Y: Cell B2:B10

- ii) Highlight A2: B10 as shown below:

	A	B	C
1	Daily Rainfall	Particulate	
2	4.1	122	
3	4.3	117	
4	5.7	112	
5	5.4	114	
6	5.9	110	
7	5	114	
8	3.6	128	
9	1.9	137	
10	7.3	104	
11			

Select Insert > Scatter >Choose first option as shown below:





ii) Using formula:

	A	B	C	D	E	F
1		Daily Rainfall(X)	Particulate(Y)	X2	Y2	XY
2		4.1	122	16.81	14884	500.2
3		4.3	117	18.49	13689	503.1
4		5.7	112	32.49	12544	638.4
5		5.4	114	29.16	12996	615.6
6		5.9	110	34.81	12100	649
7		5	114	25	12996	570
8		3.6	128	12.96	16384	460.8
9		1.9	137	3.61	18769	260.3
10		7.3	104	53.29	10816	759.2
11	Sum	43.2	1058	226.62	125178	4956.6
12	Formula used	=SUM(B2:B10)	=SUM(C2:C10)	=SUM(D2:D10	=SUM(E2:E10)	=SUM(F2:F10)

15		symbol	value	formula
16	no. of observations	(n)	9	=COUNT(B2:B10)
17				
18				
19	correlation coefficient	r	-0.978658364	=(C16*F11-B11*C11)/((SQRT(C16*D11-B11^2))*(SQRT(C16*E11-C11^2)))
20				

Using Excel Tool:

	symbol	value	formula
correlation coefficient	r	-0.978658364	=CORREL(B2:B10,C2:C10)

Assignment;

For the following information;

Age of husband	Age of wife	Age of husband	Age of wife
24	23	25	20
22	20	24	23
35	30	34	25
35	28	33	23
23	20	33	27
31	26	30	27
33	23	19	19
33	20	32	19
21	21	21	18
30	30	35	23
27	27	27	22
25	20	19	19

i) Draw scatter diagram.

ii)Interpret the calculated correlation coefficient.